CPH 100 Final Project

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Introduction

Background

Postoperative complications affect an estimated 7 to 15 percent of surgical patients, yet studying rare but clinically important outcomes (such as postoperative death or respiratory failure) requires access to large, detailed datasets [1]. Existing public resources such as MIMIC provide valuable ICU information, but their focus on critical care populations limits generalizability to the broader surgical population [2]. To advance perioperative research, external validation datasets with comprehensive ward and ICU coverage are needed.

The INSPIRE Dataset

INSPIRE ("INformative Surgical Patient dataset for Innovative Research Environment") was developed to meet this need and is openly available through PhysioNet [3]. It contains approximately 130,000 surgical cases collected between 2011 and 2020 at Seoul National University Hospital. Unlike ICU-only cohorts, INSPIRE includes perioperative data spanning operating rooms, general wards, and ICU environments [1].

The dataset records demographics, comorbidities, procedure codes, anesthesia type, intraoperative vital signs, ward and ICU vitals, laboratory tests from six months before through six months after surgery, medications, postoperative complications, length of stay, and mortality. This breadth enables analysis across the entire surgical trajectory and provides statistical power for rare outcome studies [1].

Purpose of INSPIRE and Example Application

The authors introduced INSPIRE to support collaborative research, reproducible model development, and external validation efforts [1]. To demonstrate its issue, they released an example gradient-boosted model predicting 30-day postoperative mortality, achieving an AUROC of 0.944 [1]. This example encourages benchmarking, model comparison, and exploration of clinical prediction challenges.

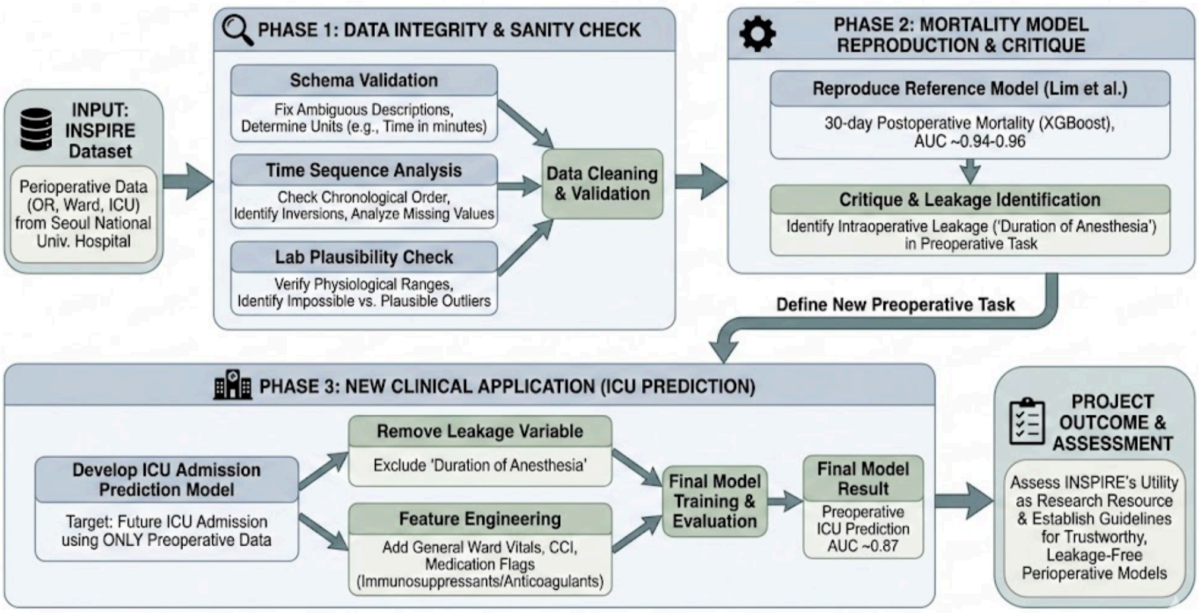
Motivation for Our Study

Our project uses INSPIRE to reproduce this mortality prediction model and critically evaluate the dataset's internal validity. Because INSPIRE includes ward as well as ICU patients, it provides a unique opportunity to examine how perioperative physiological trends contribute to risk prediction beyond critical care settings.

We are guided by three questions:

1. Is the dataset internally consistent and clinically plausible?
2. Do predictive signals reflect real preoperative information, or artifacts and potential data leakage?
3. Can the published mortality model be reproduced, and what are other clinical applications of the dataset?

By addressing these questions, we assess INSPIRE's utility as a research resource and illustrate key considerations when developing clinically trustworthy perioperative prediction models.



Data Integrity & Sanity Check

Schema

The "schema.csv" file contains the schema of every other .csv file under the INSPIRE dataset. More specifically, it contains data about the table name, column name (variable), value type, and a generic description of the column.

The first step we took in validating data integrity was ensuring that "schema.csv" contains all column names for all tables, with accurate, useful descriptions. In doing so, we've discovered some inconsistencies and ambiguities, particularly regarding:

1. Vague description for certain variables
2. Mismatching descriptions for certain variables

Exhibit A

The "height" variable in the "operations" table has an incorrect description. These type of inconsistencies pose two issues: (1) units are undefined without the correct description (2) it may be harder to know if descriptions are wrong for more complex variables

Table	Variable	Type	Description
	height	Number	Age measured before operation

Exhibit B

In the following screenshot, the "_time" based variables are ambiguous: no units are provided.

Table	Variable	Type	Description
	discharge_time	Relative Time	The time for hospital discharge
	anstart_time	Relative Time	The time for starting anaesthesia
	anend_time	Relative Time	The time for finishing anaesthesia
	cpbon_time	Relative Time	The time for starting cardiopulmonary bypass
	cpboff_time	Relative Time	The time for finishing cardiopulmonary bypass
	icuin_time	Relative Time	The time for ICU admission after surgery
	icuout_time	Relative Time	The time for ICU discharge after surgery
	inhosp_death_time	Relative Time	The time for in-hospital death
	allcause_death_time	Relative Time	The time for all-cause death

In order to determine the unit of time, we had to resort to the [original GitHub repo], look through the given code, and reverse engineer the units. Since the main prediction task is predicting if the patient will die within 30 days after surgery, we deduce that the original units are in **minutes**.

```
17    df[OUTCOME_VAR] = (df['inhosp_death_time'] < df['orout_time'] + 30 * 24 * 60)
```

A similar scenario exists with “height” and “weight” in the “operations.csv” dataset.

Exhibit C

The problem exhibited in the two following screenshots is minor, though it seems it may be easier to handle tabular data that either:

1. Groups these different drug names and atc codes into a list
2. Separate different drugs and atc codes into different row entries (alter the granularity of the "medications.csv" table)

This way, we circumvent issues with variable numbers of drug names and atc codes.

	drug_name	String	Name of the medication
	drug_name2	String	Name of the medication2
	drug_name3	String	Name of the medication3

	atc_code	String	WHO ATC code
	atc_code2	String	WHO ATC code2
	atc_code3	String	WHO ATC code3

Operations Time Entries

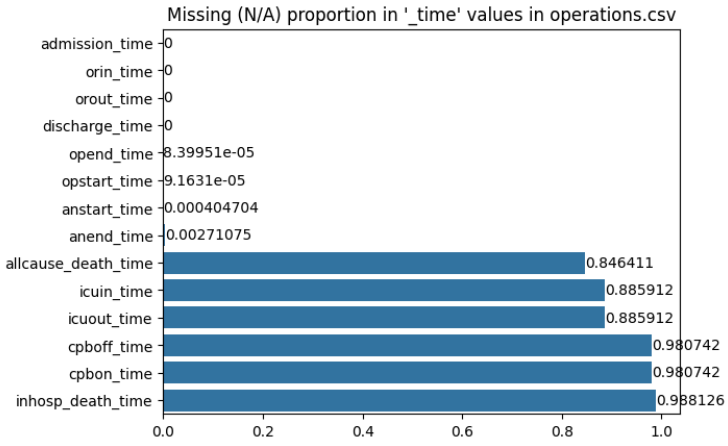
To determine whether the "_time" variables in "operations.csv" have inference value, we must ensure that (1) there isn't an abnormal abundance of missing values (2) they are ordered in the expected chronological order, with no inversions.

Missing Time Values

The following horizontal bar plot displays the proportion of NaN values in all "_time" variables, sorted ascendingly.

The most important "_time" variables that exist in any operation's record—such as hospital admission time, entering and leaving the operation room—seem to have no missing values.

Some other crucial time variables seem to have very few missing entries—such as the start and end of both the operation and anesthesia—which ideally should be none. The more concerning aspect is that the proportion of NaNs in these "time pairs" aren't equal.

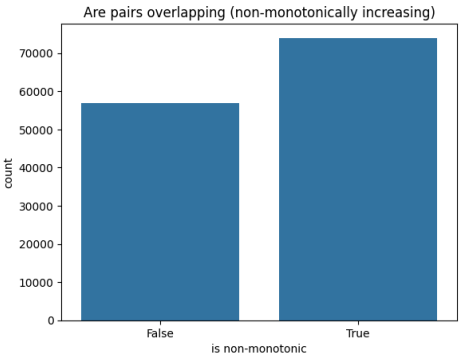


Finally, the optional time variables—such as death, ICU admission, and cardiopulmonary bypass—have large amounts of missing data, which is expected. The largest concern is the scarcity of data on operations that ended in death, particularly in the hospital, which can cause biases due to class imbalance in the original death prediction task of the paper.

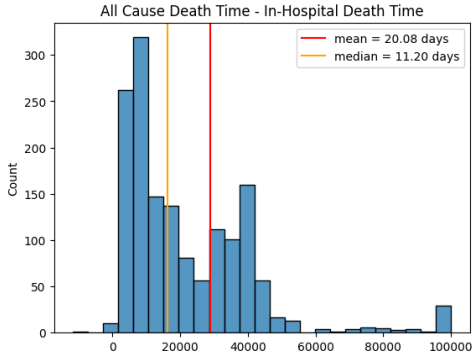
Monotonically Increasing Time?

In theory, the "required" "_time" entries must be monotonically increasing, barring "allcause_death_time" = "inhosp_death_time".

The following bar chart shows that the majority of the required time pairs include some overlapping, which is far from ideal. Similarly, the "False" bin is further inflated by considering cases where "inhosp_death_time" < "allcause_death_time".

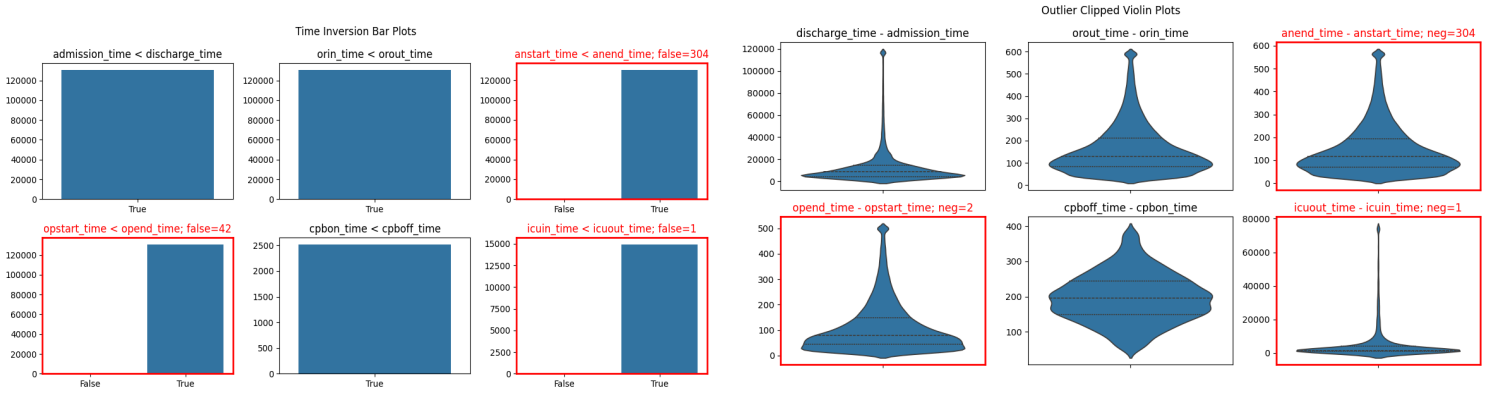


When plotting the distribution of the difference of the death times (after removing NaNs), we see a peculiar spread. In the vast majority of the cases, the "allcause_death_time" is much larger than "inhosp_death_time", with many entries being clipped at 1e5 for visualisation purposes. The mean and median of this difference is 20.08 days and 11.20 days respectively, which is a concerningly large discrepancy (before clipping).



Time-Pair Inversions & Differences

Taking a deeper look at inversions and the difference between the "end" and "start" time for all of these chronological-based pairs, we can generate the following plots:



The plots illustrate a common pattern: the majority of these time pairs have differences relatively close to 0 minutes, with very few outliers clipped at a very high difference, which could be due to special circumstances. This matches the expected distribution of these time-pairs.

The abnormal aspects are highlighted in the red outlined graphs, where the "end" time preceded the "start" time, which is impossible. Since >99% of differences in the graphs contain no inversions, we can deduce that these errors are rare and can be filtered out safely.

Lab Measurements

Measurement Range

When using lab measurements for inference and prediction tasks, it's vital to ensure that all recorded measurements are physiologically possible, and generally fall within the "typical" range.

The INSPIRE dataset contains 38 physiological measurements, in which we plotted each distribution in the grid below:

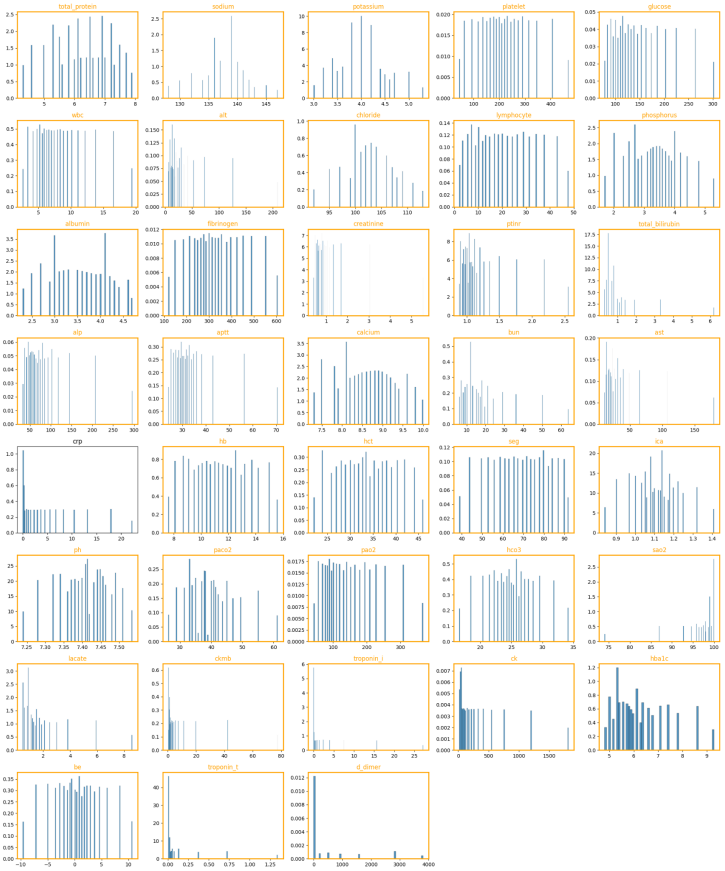
- Non-outlined histograms → both the 0.1th and 0.9th quantiles fall within "healthy" range*
- Orange outlined → either the 0.1th and 0.9th quantiles fall outside the "healthy" range, but still plausible*
- Red outlined → either the 0.1th and 0.9th quantiles contain measurements that are physiologically impossible in a living human being*

** measurement ranges are included in the python file with sources linked in the comments*

For 37/38 physiological factors, there are some entries that fall outside the healthy range, yet physiologically plausible. This is expected in a hospital record dataset, as patients can have specific illnesses and chronic conditions.

The “C-Reactive Protein” ("crp") is the only factor that has the 0.1th and 0.9th quantiles fall in the healthy range, which is useful for prediction tasks.

0/38 of the factors included impossible data, which makes the "labs.csv" table fully useful for data prediction tasks, after dropping or imputing missing measurements.



Inference & Prediction

30 Day Postoperative Mortality Classification

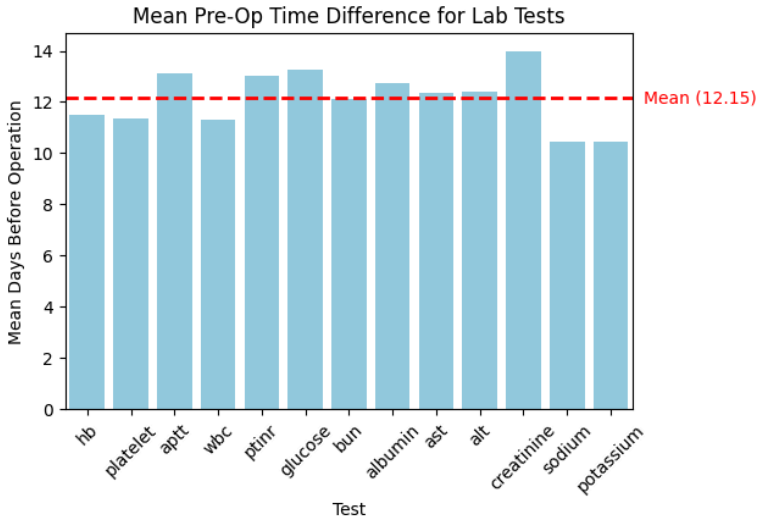
The INSPIRE dataset enables a wide variety of predictive modeling tasks due to the detailed information collected from before, during, and after surgical operations, including general ward vitals. One such classification task, as mentioned in the original INSPIRE research paper, is predicting 30-day mortality postoperation using preoperation data. Lim et al.’s demonstration model highlights the potential for different predictive models using this dataset, as it was able to achieve a 0.944 AUC using just the "operations" and "labs" tables, which is a small portion of the total data available. Their model is effective, but we aim to reproduce it, examine its limitations, and explore opportunities for improvement that better utilize the broader general ward data.

Reference Model Assumptions

Lim et al’s model makes a variety of assumptions during their pipelining and preprocessing steps due to the complicated relationship between the data tables.

First, the "operations" table contains roughly 30,000 entries corresponding to a subject's second or subsequent operations, which complicates preprocessing and integration with other tables. For example, linking a test result in "labs" to the appropriate operation becomes challenging, especially as time inconsistencies are common. To circumvent this issue, the researchers decided to reduce the dataset to each subject's first operation. In addition to this, they filter out those who are pronounced "brain-dead," which is represented by a 6 on the American Society of Anesthesiologists (ASA) physical status classification. This results in a total of 268 subjects who died (Y = 1) postoperation, and 97,992 subjects who did not (Y = 0), resulting in significant class imbalance.

Another assumption in their preprocessing pipeline occurs when lab tests are aligned with specific operations. The "labs" and "operations" tables are joined by matching each operation to the nearest preceding lab test, with a tolerance of up to six months. This means that lab values taken as far as six months before surgery are eligible to be included in the model. As shown in the figure on the right, the average time between an operation's start and the most recent corresponding lab test is approximately 12.15 days across test types.

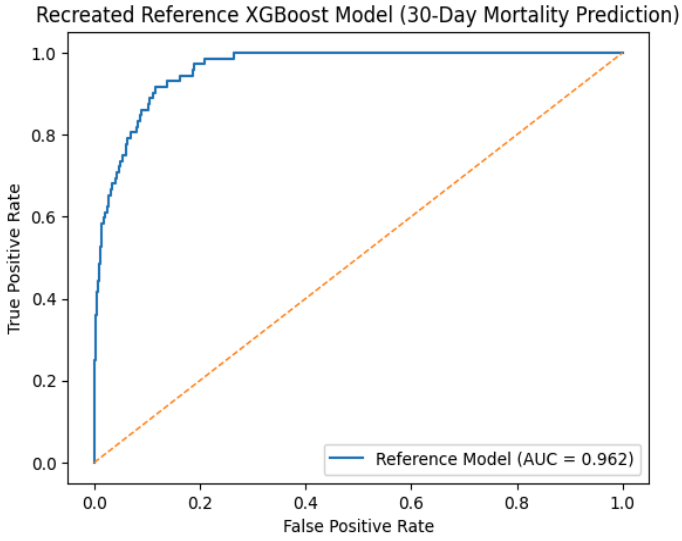


Reference Model Feature Analysis and Recreation

The authors use a wide selection of features from the "operations" and "labs" tables, including age, sex, emergency operation status, BMI, duration of anesthesia, and the previously mentioned preoperation lab results.

A feature that could affect the clinical significance of this model is the "duration of anesthesia" parameter, as its inclusion introduces interoperative data into the model. Because this parameter is likely only known after the surgery has ended, its presence may artificially boost performance if the goal is mortality prediction using preoperative data. In particular, it provides information that wouldn't normally be attainable before a surgery, such as the operation length.

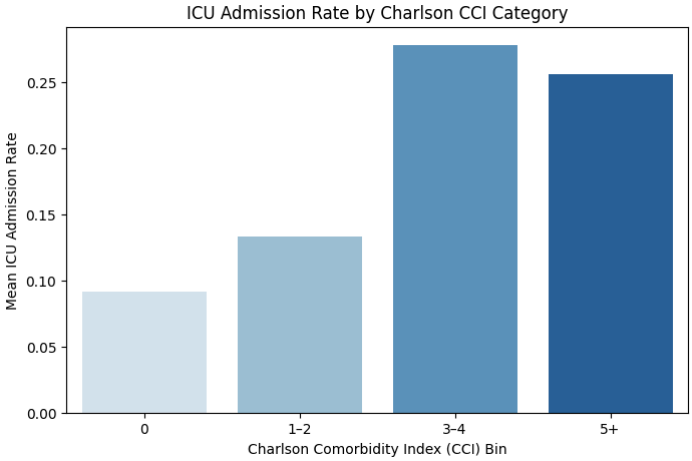
When recreating their XGBoost model in our own environment, we were able to achieve a 0.96 AUC for the prediction of death 30 days after a surgical operation.



ICU Admission Prediction Model

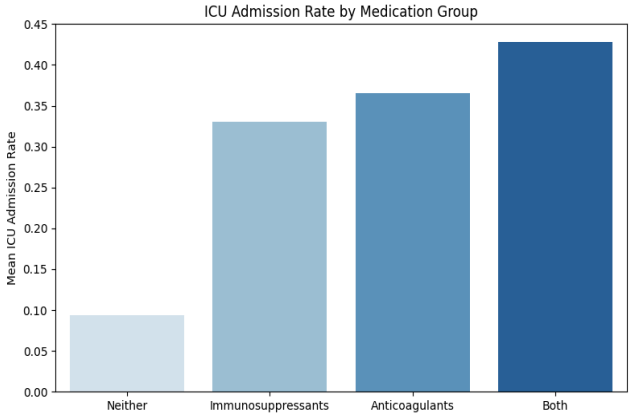
Rather than predicting 30-day mortality, our model infers future ICU admission using only preoperative data. In switching tasks, we aim to increase the model's clinical significance and impact while minimizing the class imbalance present in the previous mortality prediction task. In addition, our features include INSPIRE's general ward information by adding relevant preoperation ward vitals and diagnoses to our design matrix.

Our model uses a similar initial pipelining process, filtering to each subject's first operation, and joining based on the closest labs with a tolerance of 6 months. To ensure that our model is clinically significant for preoperative prediction, we remove the intraoperative data leakage by removing the "duration of anesthesia" feature. The removal of this feature alone greatly affected the AUC, reducing it to 0.79.



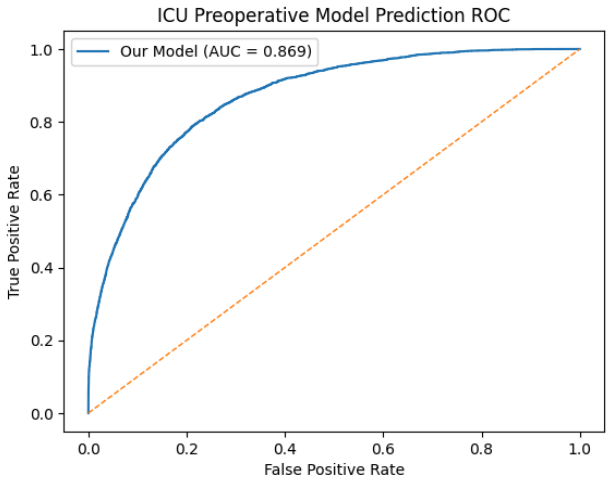
Our final model included several additional features, starting with general ward preoperative vitals such as body temperature, blood pressure, oxygen, and other core measures. Next, the Charlson Comorbidity Index (CCI) was added to the model using the "comorbidipy" library. Although this measure was originally designed as a weighted score to predict 1-year mortality based on ICD-10 comorbidities, CCI also shows a positive association with postoperative ICU admission, as seen in the figure to the left.

Our final feature addition was a flag that signaled immunosuppressive and anticoagulant medication use, which involved finding all rows associated with a patient before the start of the operation. This feature also showed a strong positive relationship with ICU admission rate, as those taking both anticoagulant and immunosuppressive medications saw heightened rates.



Results and Final Model

For the task of ICU admission prediction based on preoperative data, our final model was able to achieve an AUC of ~0.869.



References

[1] Lim, L., Lee, H., Jung, C.-W., Sim, D., Borrat, X., Pollard, T. J., Celi, L. A., Mark, R. G., Vistisen, S. T., & Lee, H.-C. (2024). INSPIRE, a publicly available research dataset for perioperative medicine. *Scientific Data*, *11*(1). https://doi.org/10.1038/s41597-024-03517-4. Accessed 9 December 2025.

[2] Johnson, A., Pollard, T., Shen, L., Lehman, L.-W., Feng, M., Ghassemi, M., Moody, B., Szolovits, P., Celi, L., & Mark, R. (2016). OPEN SUBJECT CATEGORIES Background & Summary. *MIMIC-III, a Freely Accessible Critical Care Database*. https://doi.org/10.1038/sdata.2016.35. Accessed 9 December 2025.

[3] Lim, L., & Lee, H.-C. (2024). INSPIRE, a publicly available research dataset for perioperative medicine. *PhysioNet*. https://doi.org/10.13026/46m4-f655. Accessed 9 December 2025.